

Black box approaches to activity sequence modelling

- or- *when and how does deep ML make sense for modelling human behaviour?*
- or- *do you wish (MATSim) demand generation was easier?*

Fred Shone - fredjshone@gmail.com

Dr Tim Hillel - UCL CEGE - tim.hillel@ucl.ac.uk

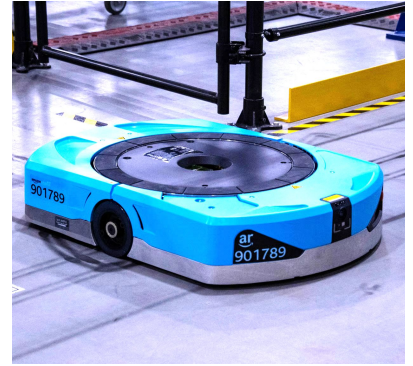
Dr Perez-Ortiz - UCL AI Centre - maria.perez@ucl.ac.uk

Contents

1. A clear case for when and how it makes sense to use deep-ML for human behavioural modelling.
2. Demonstrating how flawed LLMs are at pretending to be people(s).



How should we think of human behaviour?



Activity sequence modelling - *scheduling*

Who does what activities and when?

A critical component of all ABMs.

Difficult to model **P(schedule|labels)**;

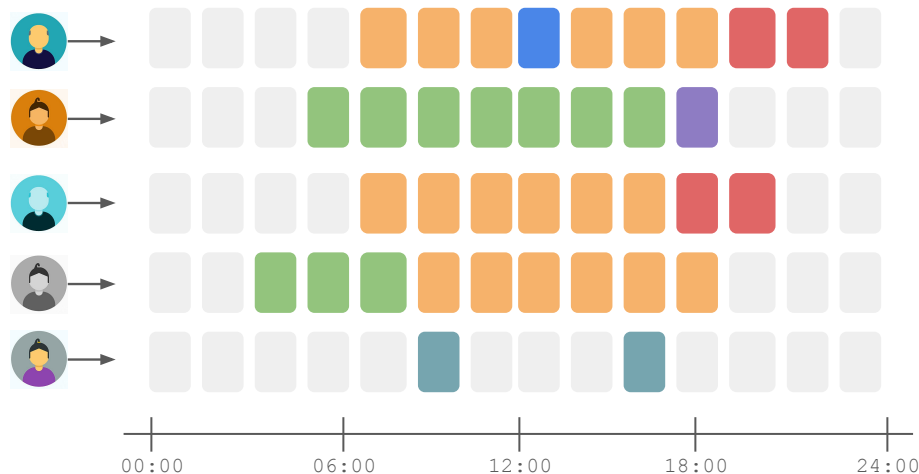
Time budget

Preferences

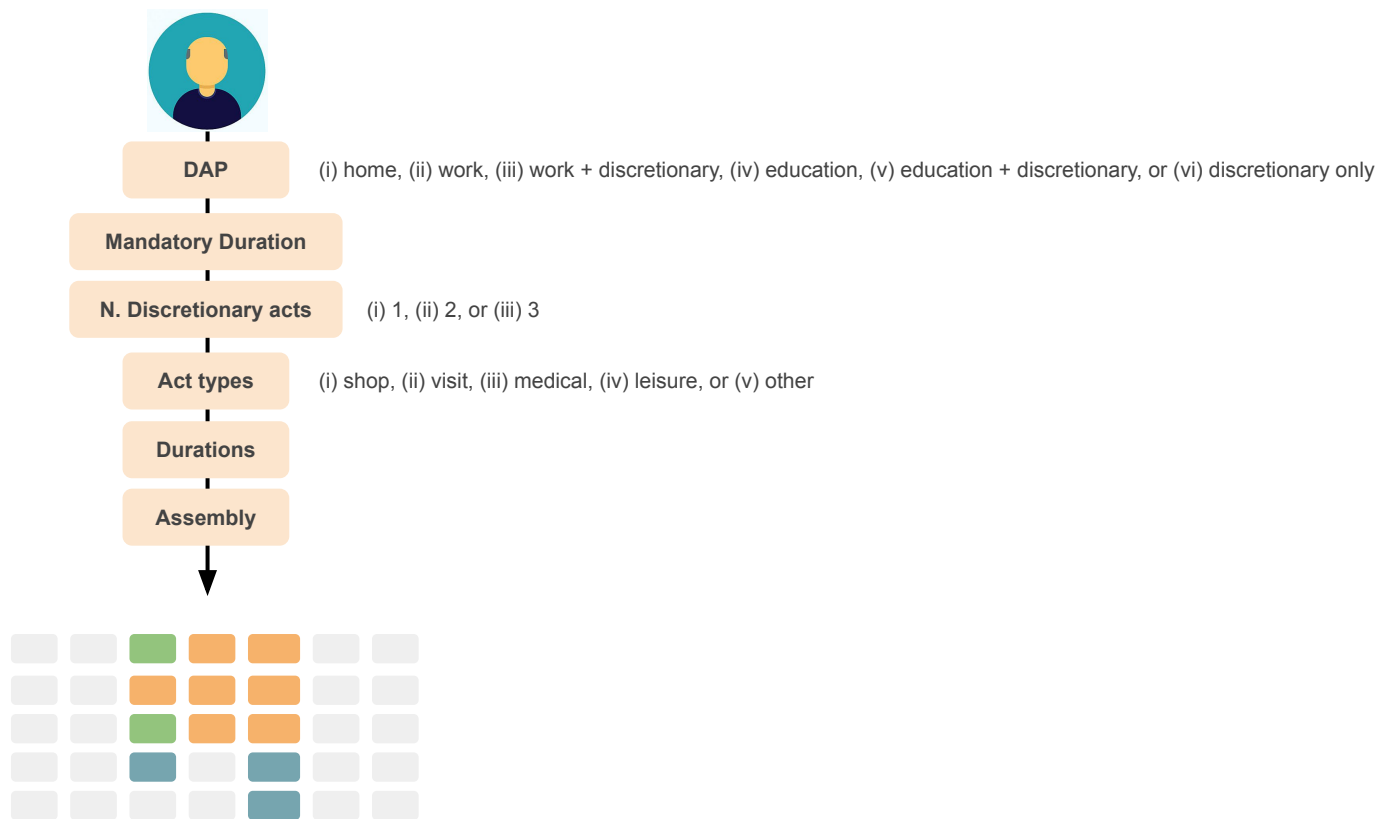
Constraints

Interactions

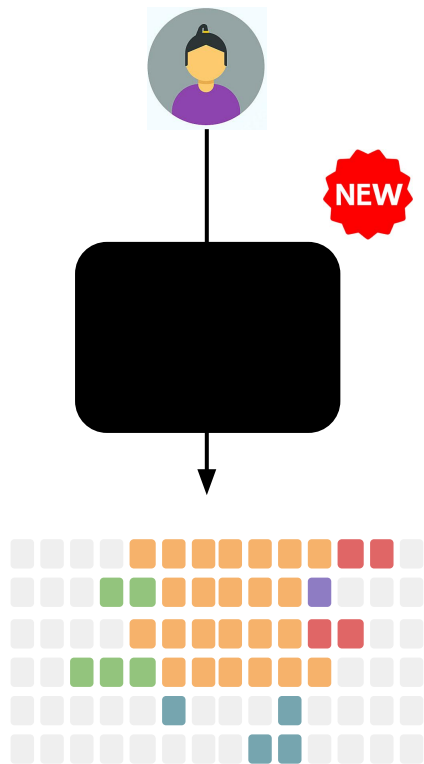
Infinite/very large choice set makes simple discrete choice setup unviable and simulation optimisation too expensive.



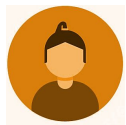
Approach **1/3**: The baseline - a *compositional** model



Approach **2/3**: Ours, ActVAE, a deep generative model



Approach **3/3**: Also, an LLM approach



You are an expert planner...

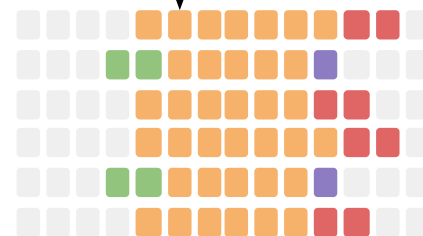
Please generate me a schedule for a 16 year old male, with no access to a car, etc...

Here are some examples of similar people...

Make sure the schedule starts and ends at home and doesn't contain consecutive home, work or education activities.

Format the schedule as json using the following specification...

```
> [{"home": 00:00}, {"work": 08:40}, [...
```



Evaluation (~60k NTS schedules):

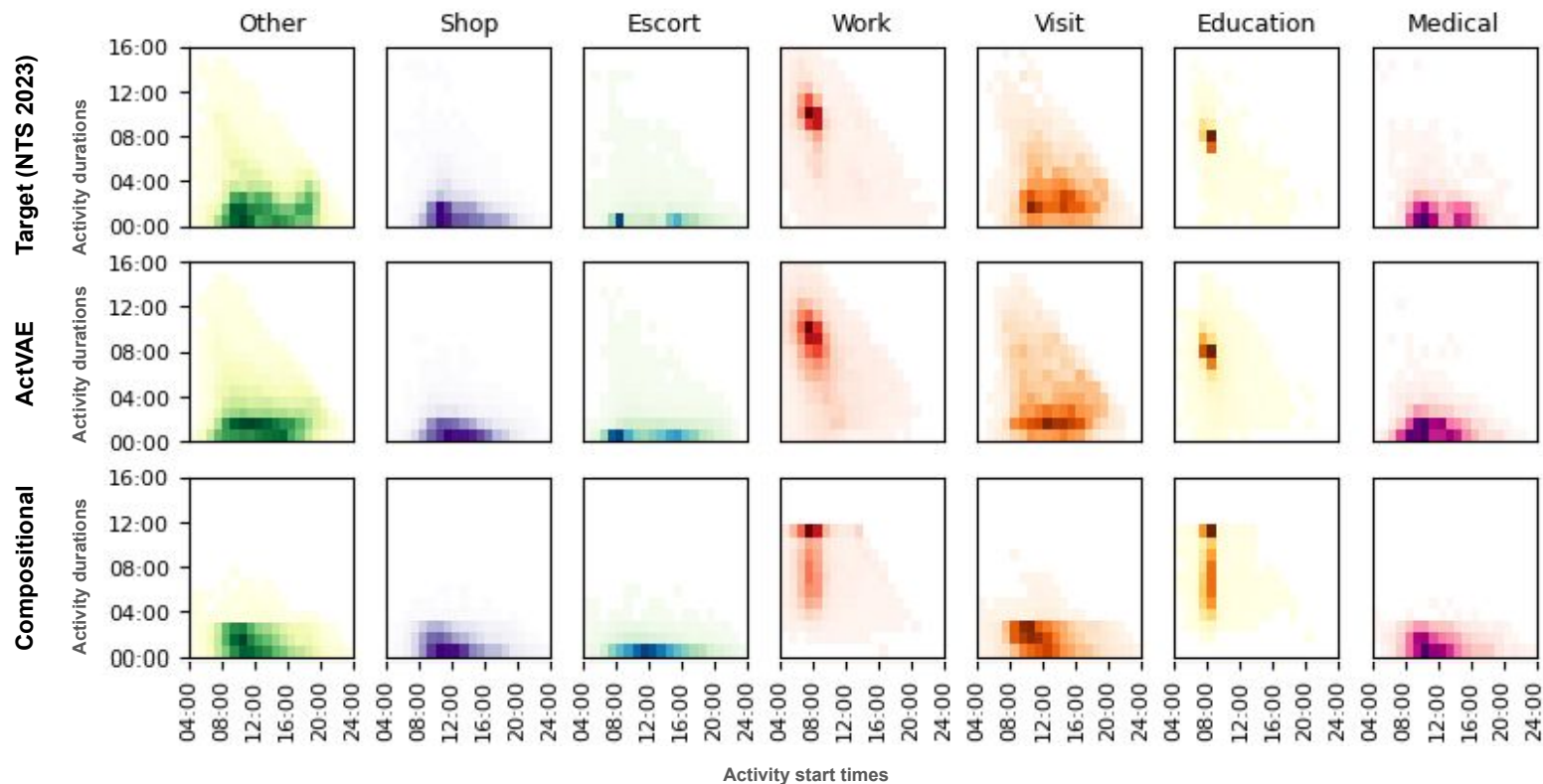


	Compositional	ActVAE	LLM	Unit
Speed/cost ↑				
Generation speed	1000	100,000	1	schedules/second
Realism ↓				
Participations distance	0.084	0.083	1.104	rate EMD
Transitions distance	0.004	0.004	0.008	rate EMD
Timing distance	0.080	0.047	0.215	days EMD
Controllability/Feasibility ↑				
Probability home-based	1.000	1.000	1.000	prob.
Non-consecutive**	1.000	0.986	1.000	prob.

* Mean from 5 model runs, variance is <0.0005 throughout.

** Adjacent home, work or education activities are considered structural zeros.

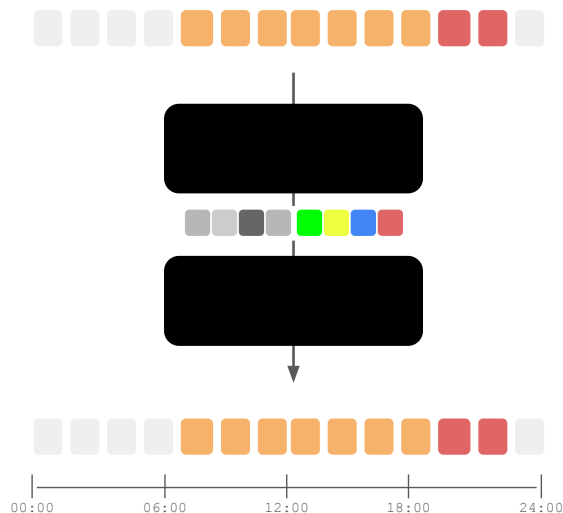
In detail: activity durations vs start times



Conclusions

- (1) When to use deep ML?
 - (a) Noisy/complex process.
 - (b) You want to go faster.
 - (c) You care less about controllability, for example for modelling system shocks, like pandemics, and more about representing diversity.
- (2) Rather than thinking of replacing whole ABM frameworks with ML, think of swapping components in where it fits. Perhaps where you don't care about interpretability.
- (3) Don't trust LLMs to behave like human(s).

Key insight 1/2: The importance of randomness



ActVAE uses a latent (compressed) representation of activity schedules (       ).

This compressed representation can be decomposed into explanatory information (age, income, etc) (   )

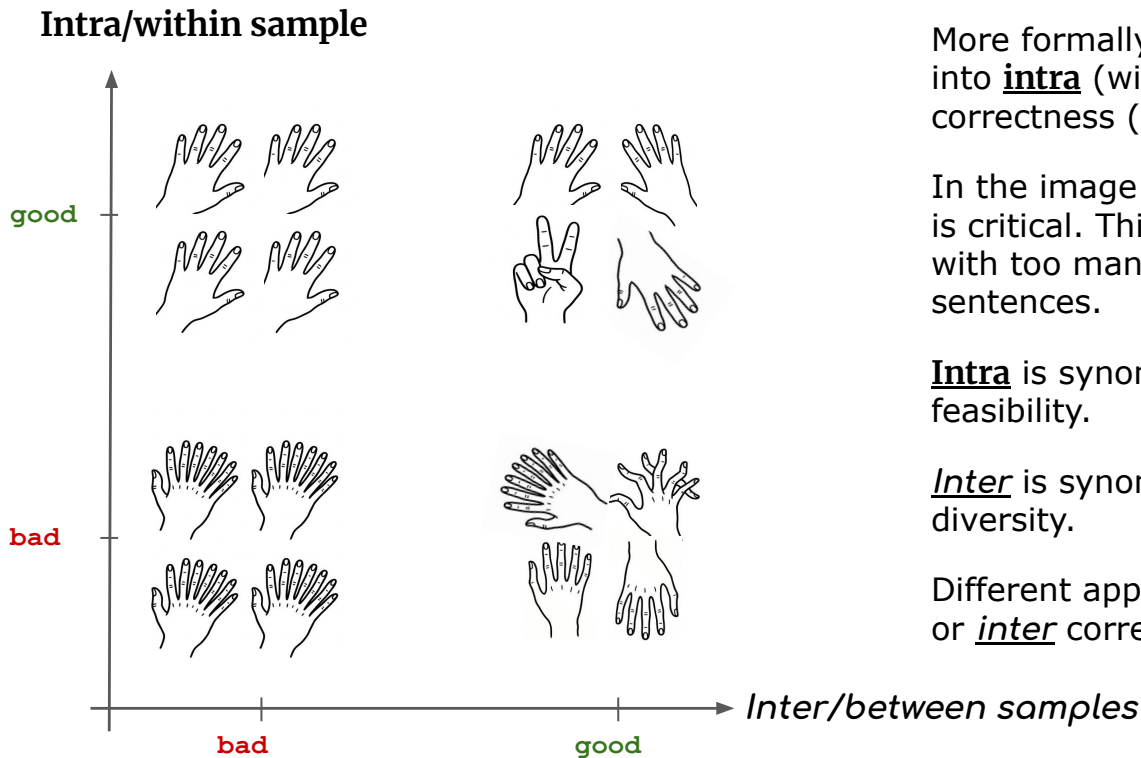
...and random variance (   ).

Using mutual information estimation we show that the random variance has ~10x more information in it than the explanatory information (    >>    ).

For scheduling (and probably many human processes we try to model) randomness dominates.

Hence the success of explicitly generative approaches - they treat unexplained variance as a first-class citizen.

Key Insight 2/2: A framework for thinking about model quality



More formally, model correctness can be decomposed into intra (within individual samples) and inter correctness (between or in aggregate).

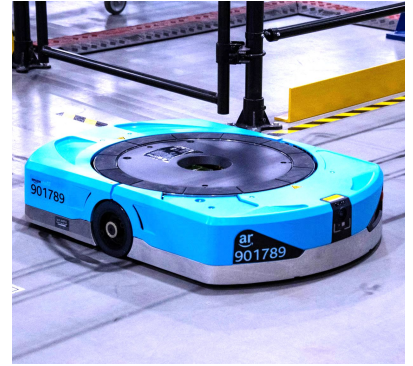
In the image and language generation domains, intra is critical. This might be avoiding images of hands with too many fingers or simply forming cohesive sentences.

Intra is synonymous with sample quality or feasibility.

Inter is synonymous with aggregate distribution or diversity.

Different approaches are biased towards either intra or inter correctness. For ABMs, **both** are important.

How should we think of human behaviour?



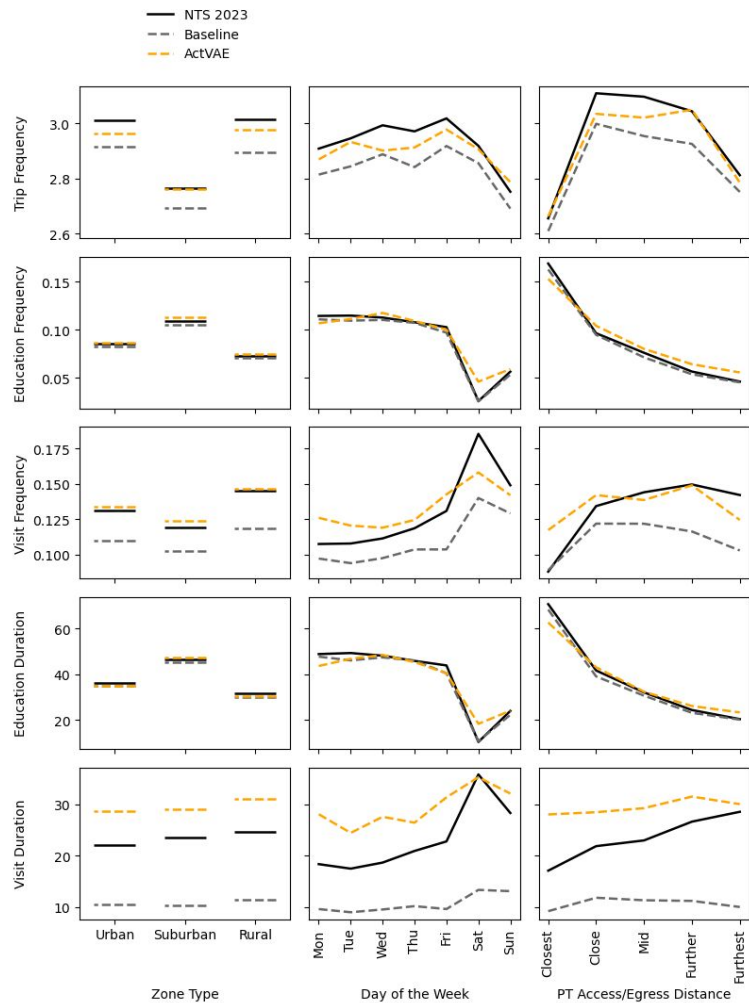
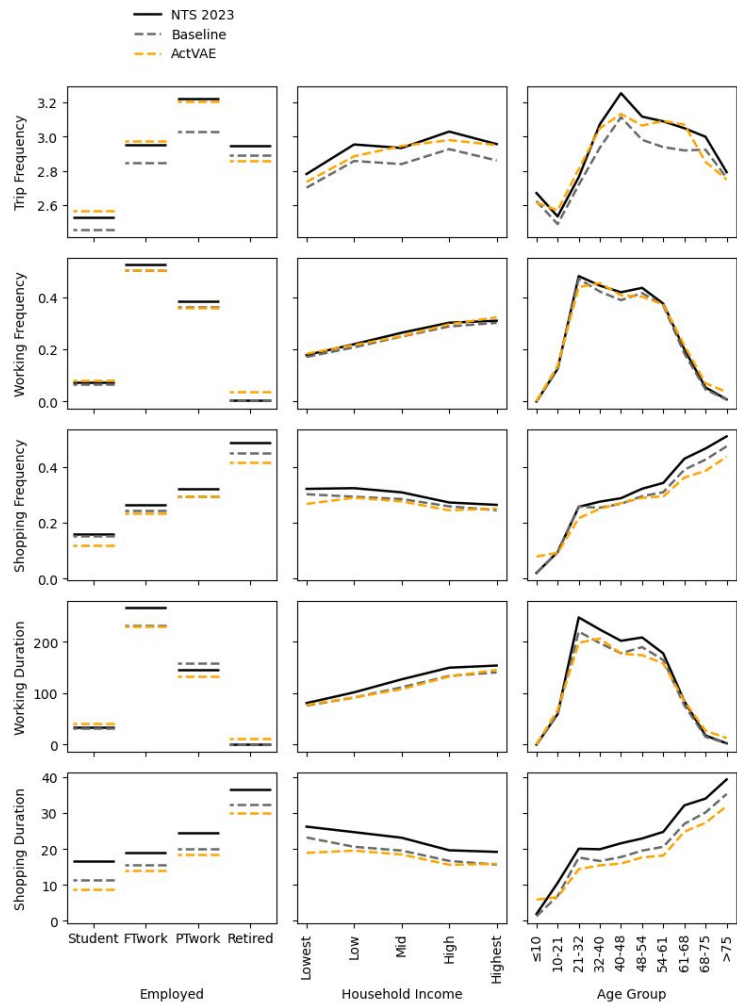
Black box approaches to activity sequence modelling

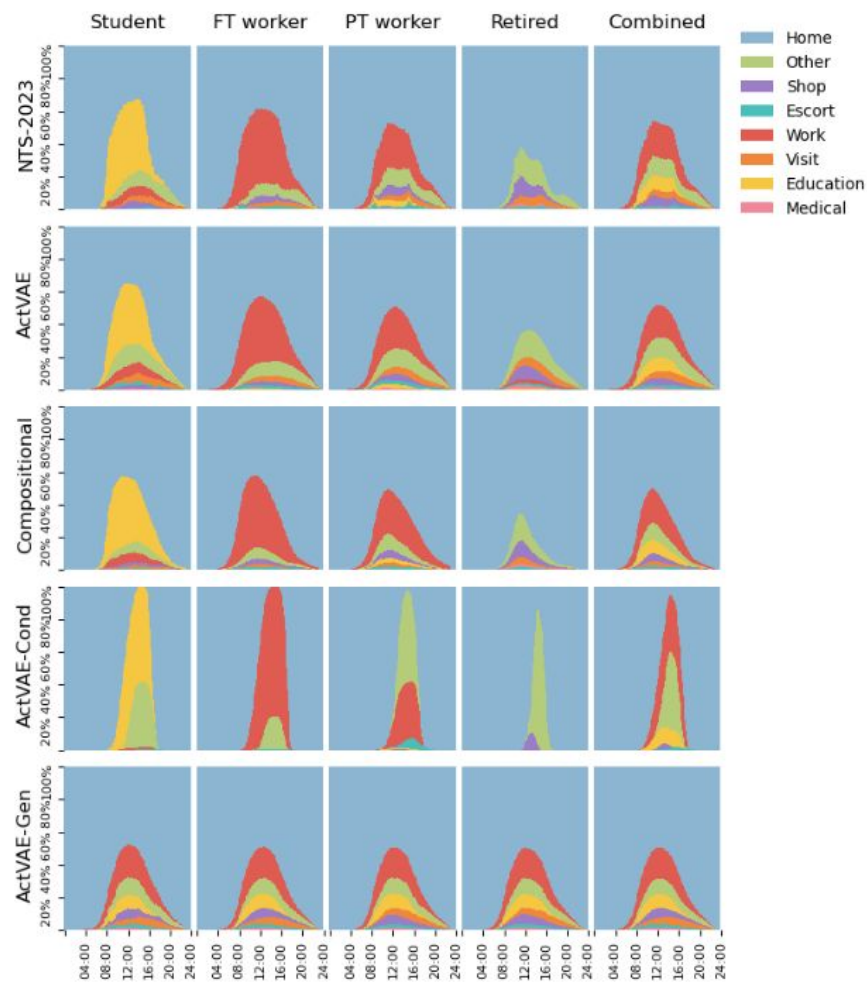
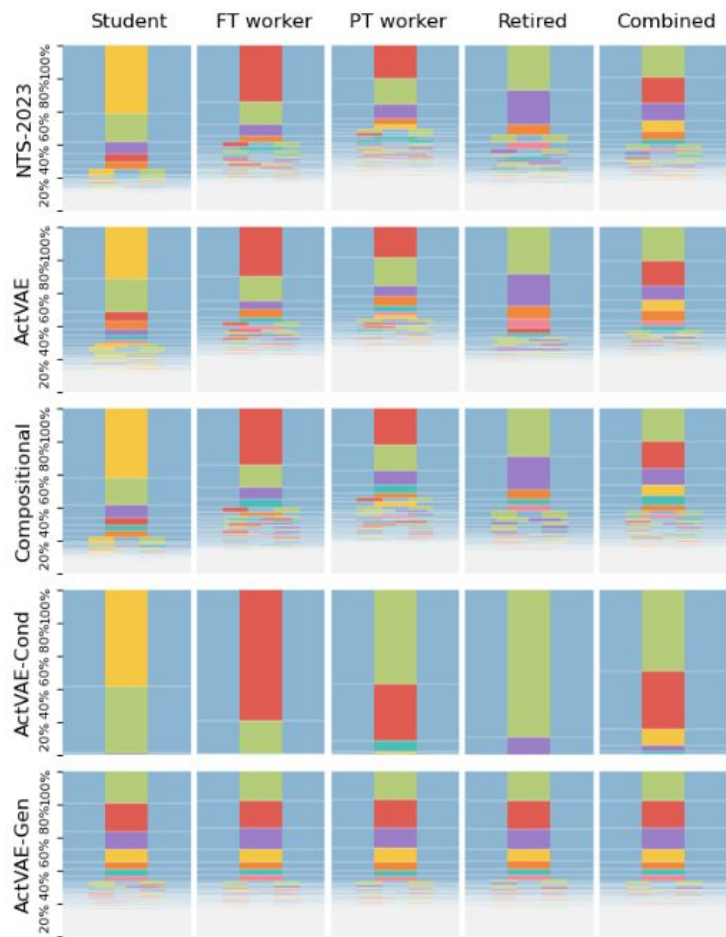
- *or, when and how does deep ML make sense for modelling human behaviour?*
- *or- do you wish (MATSim) demand generation was easier?*

Fred Shone - fredjshone@gmail.com

Dr Tim Hillel - UCL CEGE - tim.hillel@ucl.ac.uk

Dr Perez-Ortiz - UCL AI Centre - maria.perez@ucl.ac.uk





ActVAE

